

Peter Min

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Education

University of California, Berkeley: *B.S. in Mechanical Engineering*

May 2029

- GPA: 4.0/4.0
- Relevant Coursework: Calculus I–III; Linear Algebra; Physics: Mechanics and Waves; Physics: Heat, Electricity, and Magnetism; Introduction to Computer Programming; Three-Dimensional Modeling for Design; Thermodynamics; Solid Mechanics; Statistics and Data Science for Engineers

Skills

CAD: Solidworks, Fusion 360, Onshape

Programming: Python, MATLAB, C++

Experience

Undergraduate Researcher, High Performance Robotics Lab, UC Berkeley

December 2025 – Present

- PI: Prof. Mark W. Mueller
- Contributed to a team developing an autonomous vertical takeoff and landing (VTOL) fixed-wing tiltrotor system capable of lifting over 4× its own mass.
- Led manufacturing and component selection for the airframe, managing the bill of materials and ensuring the full assembly stayed within the aircraft's weight capacity.
- Managed the master CAD assembly, integrating individual components into a complete, buildable model in Fusion 360.
- Designed custom component mounts enabling secure structural connections while allowing modular assembly and rapid replacement of damaged parts.
- Sized primary load-bearing structures (spars, mounts) using hand calculations for bending stress and buckling to ensure adequate strength at minimum weight.

Propulsion and Simulations Subteam Member, Space Enterprise Berkeley

August 2025 – May 2026

- Designed a vertical lift system in Fusion 360 to elevate a horizontally resting rocket into launch position, improving testing and launch readiness efficiency.
- Supported assembly and execution of cold-flow testing to validate propulsion system performance.

Intern Researcher, California Institute of Technology

June 2024 – August 2024

- Conducted research in combinatorial graph theory with Dr. Matthew Gherman.
- Analyzed structural properties of lobster trees to identify the optimal pebbling number and position.
- Derived and formally proved a formula for the optimal pebbling number of maximal lobster trees.
- Presented findings to Caltech faculty and peer researchers.

Calculus Tutor

May 2023 – June 2024

- Worked with a student in preparation for the AP Calculus BC exam.
- Developed personalized study plans, practice tests, and targeted review problems to strengthen understanding.